

No. 194

(Dec 2025)

Cybersecurity Controls for existing ships

1. Introduction

1.1 Background

In the light of digitalization affecting shipping today, Cyber Resilience is important for "existing ships"¹ and new construction vessels, as despite their construction year, all commercially operated vessels make use, to a different extent, of certain digital systems and software prevalent in shipping. Examples of such systems² onboard are the AIS, AMS, BWMS, Cargo Handling, ECDIS and IBS used for navigation, propulsion, steering, alarm monitoring, ballast operations and several other functions.

In addition to these systems, several computer-based applications exist on board supporting functions such as ship-shore communication, planned maintenance, emissions monitoring, fleet data management, remote diagnostics, fault detection, equipment inspection, condition-based maintenance, chart updates and others.

1.2 Relation with other IACS documents on Cyber Resilience

This Recommendation focuses on existing ships and describes a minimum set of Cybersecurity Controls, to reduce the occurrence and mitigate the effects of cyber incidents, irrespective of vessel construction year. It provides a basis and does not replace any of the published IACS Unified Requirements and Recommendations on Cyber Resilience applicable to new construction vessels.

1.3 Cybersecurity Controls Periodic Review

In alignment with the evolving nature of cybersecurity threats and technological advancements, it is essential to periodically review the recommended controls to verify their continued relevance and effectiveness. It is recommended that Company incorporates periodic assessments as part of their overall cybersecurity strategy to address emerging risks and verify alignment with industry's best practices and regulatory requirements.

2. Definitions

Definitions used throughout this document are consistent with UR E26 & E27 definitions. For ease of reference, the following definitions are included.

Company: The Owner of the ship or any other organization or person such as the manager, agent or bareboat charterer, who has assumed the responsibility for operation of the ship from the shipowner and who on assuming such responsibilities has agreed to take over all the attendant duties and responsibilities.

Cyber resilience: The capability to reduce the occurrence and mitigate the effects of cyber incidents arising from the disruption or impairment of operational technology (OT) used for

¹ For this document, the term "existing ships" refers to ships contracted for construction prior to 1st July 2024.

² AIS - Automatic Identification System, AMS - Alarm Monitoring System, BWMS - Ballast Water Management System, ECDIS - Electronic Chart Display and Information System, IBS - Integrated Bridge System

No. 194

(cont)

the safe operation of a ship, which potentially lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.

Essential services: Services for propulsion, steering and safety of the ship. Essential services comprise "Primary Essential Services" and "Secondary Essential Services": Primary Essential Services are those services which need to be in continuous operation to maintain propulsion and steering; Secondary Essential Services are those services which need not necessarily be in continuous operation to maintain propulsion and steering but which are necessary for maintaining the vessel's safety.

Firewall: A logical or physical barrier that monitors and controls incoming and outgoing network traffic controlled via predefined rules.

Information Technology (IT): Devices, software and associated networking focusing on the use of data as information, as opposed to Operational Technology (OT).

Network: A connection between two or more computers for the purpose of communicating data electronically by means of agreed communication protocols.

Network segment: In the context of this recommendation, a network segment is an OSI layer-2 Ethernet segment (a broadcast domain). Note on TCP/IP: Network address plan is prefixed by their IP addresses and the network mask. Communication between network segments is only possible by the use of routing service at network layer (OSI Layer 3).

Operational Technology (OT): Devices, sensors, software and associated networking that monitor and control onboard systems. Operational technology systems may be thought of as focusing on the use of data to control or monitor physical processes.

3. Scope and selection criteria

3.1 Recommendation scope

The selected Cybersecurity Controls do not only consider the technologies used in shipping but also take into consideration that ships are in operation.

While this Recommendation is primarily directed at shipping companies, it is also relevant to other stakeholders involved in implementing cybersecurity controls—such as vessel owners, operators, and ship management companies. The focus of this document includes specific vessel types and systems supporting essential services onboard.

3.2 Vessels

This Recommendation is applicable to all vessels over 100 GT, whether self-propelled or not, operating in restricted or unrestricted service. However, it excludes "inland waterway" vessels, military vessels, and other government ships operated for non-commercial purposes.³

3.3 Systems

It is assumed that existing ships would have been provided with IT systems and OT systems. Since both IT systems and OT systems are potentially vulnerable to cyber threats, both IT systems and OT systems should be protected. For the purposes of this IACS

³ IACS Procedures, Volume 1, Rev. 19 June 2024

No. 194

(cont)

Recommendation, the primary focus is on OT systems, as referenced in IACS UR E26 1.3.2(a), and their possible connections to IT systems, as also referenced in IACS UR E26 1.3.2 (b).

Some existing ships may have been originally constructed without taking into consideration concerns related to interconnections of IT systems with OT systems. For such ships, IACS recommends separation of the OT systems from the IT systems. Please see the row for Item No. 2 “Network Protection” in the below Table which indicates that firewalls should be supplied to provide network segmentation (with at least three zones, the OT zone, the IT zone and the crew zone). Without such separation the OT systems may be exposed to additional cyber threats via the IT systems.

3.4 Selection Criteria (for controls)

The selection criteria for the recommended Cybersecurity Controls relate to requirements described in IACS UR E26 and technologies available and used by shipping companies today. Onboard procedures are also taken into consideration.

The following criteria have been considered during Controls selection:

- Limited in number
- Technology agnostic, allowing the Company to choose from available options
- Cause minimal technical and procedural implementation effort
- Have wide application regardless of ship type and years in service
- Ships indicative ages⁴ for implementation:
 - Passengers ships up to 35 years in service
 - Tankers, Bulkers, Containers, Service, Tugs, etc. up to 20 years in service

4. Controls

4.1 System Types, Description

The following table includes 4 columns:

- Item No: used for sequential numbering.
- Controls/Procedure: Stating the technical and/or procedural controls, applicable to OT system types, as indicated in each control and a reference to the corresponding UR E26 requirement.
- Rational: Explaining the reason each control is proposed, with reference to IACS UR E26 rational when applicable.
- Description: Providing further information for each control

⁴It is common practice in shipping, that even vessels of this age, have received system upgrades and retrofits, due to operational requirements, regulatory compliance, business reasons, or other.

No. 194

(cont)

4.2 Cybersecurity Controls - Table

Item No	Controls/Procedure	Rationale	Description
IDENTIFY			
1	Vessel Asset Inventory <i>(UR E26 Reference par. 4.1.1; IACS Recommendation No. 190 on Vessel Asset Inventory for Computer-Based Systems)</i>	Rationale: See UR E26, para. 4.1.1.2 Rec. No. 190	Lists of OT systems onboard. These may include: a) manual lists, e.g. Excel spreadsheets b) software generated e.g. from an asset scanner c) combination of both manual and automated sources d) description of each system and key attributes such as type of equipment, manufacturer, software version, patch update
PROTECT			
2	Network protection <i>(UR E26 Reference par. 4.2.1 & 4.2.2)</i>	Rationale: See UR E26, para. 4.2.1.2 and 4.2.2.2	The ship OT systems should be segmented from other zones by firewalls or other security devices capable of providing (refer to section 3.3): a) network traffic monitoring b) real time alerting c) IT/OT segmentation d) Crew zone segmentation e) micro-OT (east-west) segmentation, where applicable
3	Antivirus <i>(UR E26 Reference par. 4.2.3)</i>	Rationale: See UR E26, para. 4.2.3.2	Antivirus, antimalware, and/or other protections from malicious code. For OT systems Antivirus protection can be used only if allowed or provided by the system supplier.
4	Access control <i>(UR E26 Reference par. 4.2.4)</i>	Rationale: See UR E26, Para 4.2.4.2	Covering the following: a) Physical access security to systems supporting essential services onboard. b) Logical access control (when used or applicable). c) Digital identity & access management measures (when used or applicable), such as Role Based Access. d) Management of credentials (when used or applicable).
5	Wireless communication <i>(UR E26 Reference par. 4.2.5)</i>	Rationale: See UR E26, Para 4.2.5.2	In case of wireless communication onboard the following apply: a) Should be the least preferred method for OT. b) No primary essential services should be connected. (IACS E22 cat III categorization). c) Wireless devices should be in segregated security zones. d) Only company authorized processes and devices should be allowed to communicate on the wireless network.
6	Remote access control <i>(UR E26 Reference par. 4.2.6)</i>	Rationale: See UR E26, Para 4.2.6.2	Recommended practice for remote access to onboard systems: a) multifactor authentication of the remote user ashore. b) limit of unsuccessful login attempts. c) connections are explicitly accepted by responsible personnel on board & managed by personnel onshore.

No. 194

(cont)

			d) can be manually terminated by personnel on board or automatically terminated after a period of inactivity. e) use of strong encryption algorithms and protocols to ensure the confidentiality, integrity, and authenticity of the data being transmitted. f) Activity log of remote sessions.
7	USB protection / use of portable devices <i>(UR E26 Reference par. 4.2.7)</i>	Rationale: See UR E26, Para 4.2.7.2	a) Block unused ports by physical plugs and maintain keys with authorized personnel. b) Software protection if or when provided by the supplier. The portable device can be scanned by external antivirus software with up-to-date antivirus signatures.
8	Training <i>(UR E26 Reference par. 5.3)</i>	Rationale: See UR E26, Para.5.3 "The shipowner shall prepare and implement operational procedures, provide periodic training and carry out drills"	Familiarization & training for Crew in relation to OT related cyber risks malfunctions & procedures.
9	Systems updates & maintenance <i>(UR E26 Reference par. 4.1.1.4.4)</i>	Rationale: See UR E26, Par.4.1.1.3 "The inventory shall be kept updated during the entire life of the ship"	Systems updates/ maintenance activities by company, supplier or 3rd party on pre agreed frequency.⁵ Systems updates/ maintenance by qualified, authorized personnel. Available change management records.
DETECT			
10	Network Operation Monitoring <i>(UR E26 Reference par. 4.3.1)</i>	Rationale: See UR E26, Para.4.3.1.2	Should have adequate monitoring, diagnostic, and logging functionalities enabling the detection of anomalies.
RESPOND			
11	Responsible person(s) ashore <i>(UR E26 Reference par. 4.5.1.3)</i>	Rationale: See UR E26, Para.4.4.1.2 "Designated personnel ashore should be integrated with the ship in the event of a cyber incident"	Company Point/s of Contact (PoC) ashore to support the ship master & crew in case of cyber incident or malfunction.
12	Reporting <i>(UR E26 Reference par. 4.4.1.3)</i>	Rationale: See UR E26, Para.4.4.1.3 "The Incident response plan shall provide procedures to respond..."	Company procedures and reporting form/s for cyber incident or malfunction.
RECOVER			
13	Backup & systems restore <i>(UR E26 Reference par. 4.5.2)</i>	Rationale: See UR E26, Para 4.5.2.2	For OT systems (one or both of the below options): a) Duplication of systems and equipment as required by statutory regulations of flag state government or class rules. b) Backup & restore procedures for other digital systems (provided by the supplier or developed by the company).
14	Incident Detection, Response & Recovery	Rationale: See UR E26, Para.4.4.1.3 "The Incident response plan shall provide procedures to respond..."	Company/Supplier procedures to detect and respond to systems anomalies/errors or Incident Detection Response & Recovery actions (including Network Isolation procedures).

⁵e.g., hourly/on-call/ monthly/yearly etc., depending on the equipment

No.
194
(cont)

	(UR E26 Reference par. 4.4.1 & 4.5.1)	Par.4.3.1.2 “The Monitoring system that can...” and Para 4.4.3.2	
--	---------------------------------------	--	--

End of
Document